CHAPTER 11

CLEAN SOURCES OF ENERGY

Some will claim that limited resources prevent us from achieving a society of abundance. This is simply not so. We still have more than enough resources to achieve a high standard of living for everyone. But it's time to move beyond our failed programs and frustrations to innovative solutions that could be readily applied if we were to direct our attention to overcoming scarcity. We have the capability to intelligently apply the humane use of science and new technology to provide for most human needs, and to reclaim and restore the natural environment.

Fossil fuels such as oil and coal allowed our civilization to progress to its present state of development. However, these energy sources are also limited and non-renewable resources, and one of the many environmental dangers.

In the process of designing a new civilization we must harness energy, a major source of the material well-being for all nations. This is a double-edged sword. When placed in the hands of private interest and greed, energy can be used for destruction. The current stock of atomic weapons can destroy the world many times over. But fusion power and other forms of clean sources of energy, when applied intelligently with emphasis on human and environmental concern, could provide all of the nations of the world with clean, unlimited energy sources and a standard of living unattainable today.

Much remains to be accomplished in the undeveloped areas of our planet. Vast and untapped energy sources remain largely unexplored and untouched; amongst these are wind, wave and tidal action, ocean currents, deep ocean pressure and temperature differentials, falling water, geothermal and electrostatic power, hydrogen and natural gas, algae, bacteria, phase transformation, and thermionics, or the conversion of heat into electricity by boiling electrons off a hot metal surface and condensing them on a cooler surface. Additionally, there is the untapped potential of Fresnel lenses, inflatable dome versions of which are being developed for use as optical concentrators in solar power systems.

Fusion power welds together light atoms such as hydrogen and lithium. Fusion energy is the same energy that drives the cosmos and the stars. When we learn how to harness it, the world's energy problems can be solved forever, without any detrimental effects or dangerous toxic materials to be disposed of. The only residue would be the clean ash of helium.

Oceanographers tell us that the world's oceans, occupying 70.8% of the earth's surface, possess an endless supply of surging energy called deuterium, a heavy hydrogen atom locked in the seawaters. According to John D. Isaacs and Walter R. Schmitt, the amount of fissionable uranium and thorium in the oceans can support our present level of power production for millions of years. It is highly probable that in the next century our main source of energy will be produced by thermonuclear fusion or geothermal extraction. Both appear relatively free of many of the hazards inherent in energy produced by nuclear fission.

The transfer of electrical energy will probably be facilitated by improved methods of superconductivity, utilizing cryogenics as part of the international power grid. This grid could serve primarily as a supplement or backup to self-generating structures within the cities. A key element of the design of cities in the future will be the embedding of all necessary energy creation within the structure of the city itself.

We could also utilize solar concentrators as an alternative to fossil fuels for the generation of heat. As of this writing, the Argonne National Laboratory and ARDI are developing a production technique for solar cells that will be nearly 70% efficient at a cost one-tenth that of silicone-based cells. There are many other possibilities for developing photovoltaic systems that generate electricity while harnessing the currently untapped radiant heat energy.

The world's most powerful single hydro-electric project is now being constructed in the Tsangbo Bend in Eastern Tibet, where the river Tsangbo is fed by the great glaciers and waterfalls which descend over seven thousand feet. When the Chinese harness the energy of this dam, it is estimated that the turbines in this power project will be capable of producing far greater than forty million horsepower. This is equal to the total world production of hydroelectric power today.

Another vast, untapped energy option is the development of piezoelectric materials. This source could be employed by using laminated systems inside cylinders, activated by the rise and fall of the tides. A recent development of one these materials is polyvinylidene-fluoride. Five square kilometers can supply electricity for two-hundred-and-fifty thousand people at a cost of one to three cents per kilowatt power, a considerable savings over fossil fuels.

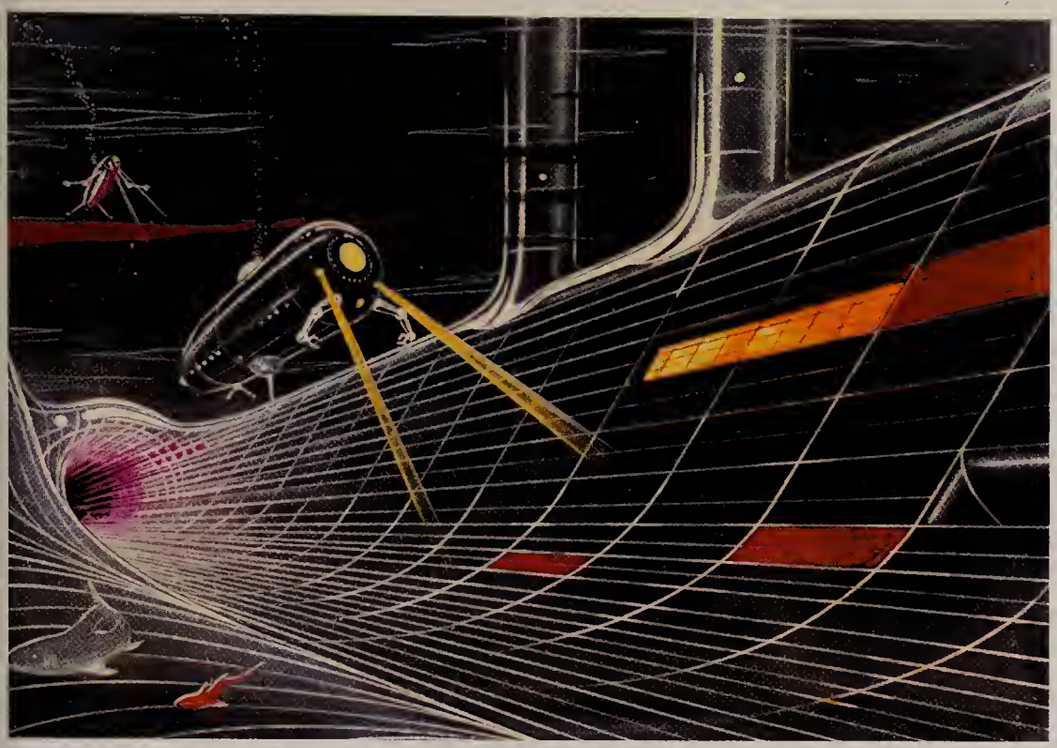
If we developed and harnessed only 1% of the geothermal energy that is available in the crust of the earth, all of our energy problems would be eliminated. Geothermal energy can supply us with more than 500 times the energy contained in all the fossil, oil, and gas resources in the world. Geothermal power plants produce very little sulfur, compared to fossil fuels, and emit no nitrogen oxide or carbon dioxide. A relatively small area of land is required for the power plant itself. Geothermal power is the most economical and efficient way to heat and cool buildings. Natural heat stored underground in combination with the permafrost zones could generate thermal electric power and utilize this power to cool buildings in warm weather by means of geothermal heat pumps.

Geothermal energy can also be used to grow plants year round in enclosed areas, as has already been accomplished in Iceland and other areas. In this way an enormous amount of fresh vegetables could be cultivated in all seasons. A similar process could be used for fish farming and all other regions where heating and cooling is needed. If we had applied just one tenth of what we've spent on military equipment to the development of geothermal generators, we could have long ago solved any energy shortages.

In our resource-based economy, a comprehensive analysis of the environmental, human, and social impact would be carefully analyzed before construction on any project. In all endeavors, the major concern would be to protect and restore the environment for the benefit of all living creatures in the community of life. Of course, the purpose of the construction and development of these power projects is also to free human beings from unnecessary expenditure of energy and from laborious tasks. Up to the present, social development in our monetary oriented society evolved in a haphazard manner, affected by many interacting variables. This process only retarded the advantages inherent in a global cooperative project to develop renewable energy resources rather than exhaust limited resources.

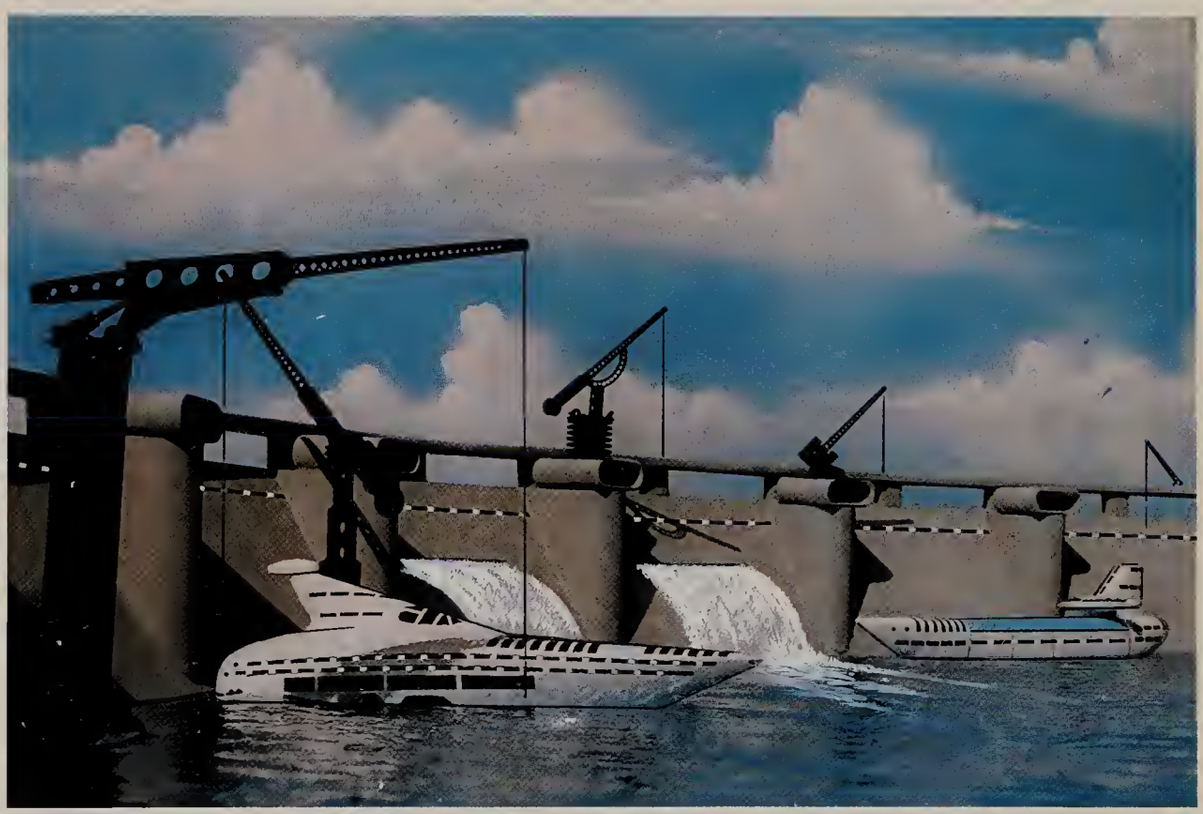
At present we have the means to determine globally the best energy resources available for each geographical location on our planet. What is desperately needed in this world of high technology and rapid change is an energy development strategy on a global scale. To develop a global, sustainable strategy would call for a joint venture of international planning on a level never before achieved.

Eventually international life arteries could serve all nations economically and efficiently. Only by utilizing the best planetary planning can wasteful consumption be reduced considerably. And it is only by reducing wasteful consumption that we can achieve our end goal, that of the highest possible standard of living for all of the world's peoples.



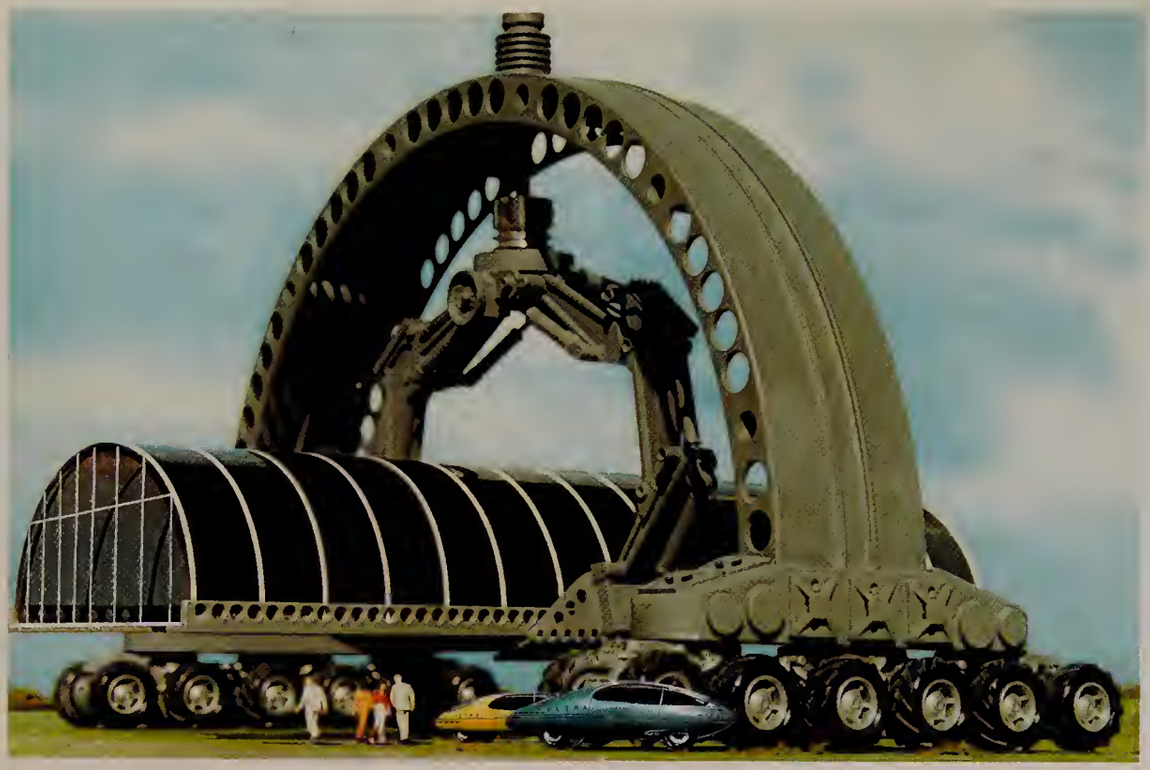
HARNESSING THE GULF STREAM

These underwater structures convert a portion of the flow of the Gulf Stream through turbines to generate clean electric power. The turbines would have a centrifugal separator and deflectors to prevent harm to marine life.



BERING STRAIT DAM

A major development in the future could be the construction of a land bridge or tunnel across the Bering Strait. The primary function of this span would be to generate electrical power and house facilities for collecting and processing marine products. Beneath and above the ocean surface would be tunnels to transport passengers and materials. Pipelines to conduct fresh water from melting icebergs to other parts of the world may also be incorporated. Not only could this structure provide a physical link between Asia and North America, it could also serve as an avenue for social and cultural exchange.



DESALINIZATION PLANT

This mega-machine transports a transparent enclosure used for evaporative condensation. It would be placed over canals, some of them containing salt water, and could serve as a desalinization plant to supply clean water for drinking, irrigation, and other needs. This is accomplished by harnessing the power of the sun and will eliminate water shortages throughout the world.



GEOTHERMAL ENERGY PLANTS

In the future, as refinements in conversion technologies increase its feasibility, geothermal energy will take a more prominent role in reducing the threat of global warming. Readily available in many regions throughout the world, this source alone could provide enough clean energy for the next thousand years.

CHAPTER 12

CHANGING HUMAN NATURE

MUCH OF THE BEHAVIOR ACCEPTABLE TODAY would be socially offensive in a saner or more logical social arrangement. But whatever values, ideals, and behavior people aspire to cannot be fully realized when there is hunger, unemployment, deprivation, war, and poverty. People deprived of their income will often do whatever is necessary to provide the necessities of life for themselves and their families. Their values may be exemplary, but their behavior will reflect the reality of the situation. After World War II, for example, even the most respectable German families could be seen fighting over scraps of food in garbage cans to survive. In a scarcity-oriented society generosity is often a rare occurrence.

It is not enough to design new cities and make sweeping generalizations about human participation and democratic ideals. We must reexamine our dominant values and how and why they evolved.

During the transition to this saner world, there will still be a great deal of interpersonal conflict, egocentric behavior, and all of the other problems that plague our present day monetary society. Therefore it is essential that we utilize newer methods of evaluation to greatly enhance human behavior.

When we examine human behavior in the same manner as any other physical phenomenon, we will be more capable of understanding the physical factors that are responsible for shaping our values and behavior. In the natural sciences all physical phenomena are acted upon by resident forces. For example, a sailboat does not sail of its own accord; rather, it is activated by the wind. A telephone pole does not just fall to the ground; it is acted upon by rain, gravity, wind and a number of other variables.

Human behavior in all areas is just as subject to natural laws and the actions of external forces: it is generated by the many interacting variables in one's environment. This applies even to behavior that is socially offensive. It is often influenced either by one's experiential back-ground, nutritional factors in early life, or a number of other interrelated environmental factors. When one sees a dog leading a blind person across the street we tend to think it is a good dog. But when we see a dog behave offensively we call it a bad dog. The dog is neither good nor bad. A dog can be trained to be ferocious or to help the blind. Both animals could be of the same breed, even from the same litter. Their behavior is due to the differences in training.

To put it another way, try to imagine an ancient Roman family watching Christians being fed to lions. Someone of today might be horrified and believe the people watching had trouble sleeping that evening. But they most likely had no trouble sleeping at all. Such bloodshed was the cultural sport of the times. Lion and Christian were looked upon with equal disdain.

Or imagine a modern day fighter pilot, trained in warfare and taught a similar disregard for the other's culture and beliefs, losing sleep over the fact that he shot down twenty planes and burned several inhabited villages. More likely, he will beam as he is presented a medal, and adorn his aircraft with symbols of his "kills." The pilot reflects the values of his culture just as the Roman family does. What we call our 'conscience' and 'morality' are not determined by an invisible "higher self". They are largely determined by geography, the times, and the individual's upbringing.

One of the greatest limiting factors in human systems is our inability to grasp the significance of resident forces and the extent to which that environment shapes our thinking, values, and/or behavior. When we speak of environment we mean all of the interacting variables, which are the prime contributors of our mindset.

A fundamental consideration in the physical sciences is that one must attempt to identify all of the physical factors responsible for certain outcomes. For example, when an automobile acts in an unusual way most mechanics can account for the reasons and identify the physical factors for that condition. When a human being appears at a hospital with a certain kind of injury, even if he or she is unconscious and unable to identify the cause of the injury a competent medical staff can usually identify the probable cause.

With certain forms of aberrant behavior neurologists, biochemists and psychiatrists can, to a limited extent, identify some of the conditions responsible for this behavior. Even in everyday life, evidence supports the connectivity of influential events all around us. But we often fail to apply the same methods of evaluation that are used in the physical sciences to human behavior.

In many instances our collective values are influenced by an existing social structure or sub-culture within society. For ill or well, social systems generally tend to perpetuate themselves and all of their strengths and shortcomings. In our era of mass-communication, the media-controllers and established institutions influence the national "agenda," which in turn influences much of our behavior, expectations and values.

Whether they realize it or not, most people are constantly manipulated through the media. If you doubt this, check your public TV station for an international news broadcast. Comparing that newscast to you regular newscast could easily lead you to think that the reporting originated from different planets. One must watch with great skepticism.

Our most cherished beliefs are influenced by books, motion pictures, television, religions, role models and the environment we live in. Even our notions of good and evil and our concepts of morality are part of our cultural heritage and experiences. This method of

control does not require the use of physical force and has been so successful that we no longer recognize or feel the manipulation.

The dominant values of any social system rarely come from the people. Rather, they represent the views of the dominant control group — the church, the military, the banks, the power elite or any combination of these — which determines the public agenda, the courts, taxes, etc., all of which serve their own interests and perpetuate the illusion that society's values are determined from the ground up. And governments suppress or explain away any deviations that may threaten them.

The fear of retribution from gods and demons is still effective at controlling ignorant and superstitious populations in both developed and undeveloped nations. There are many who actually believe that demons are responsible for anti-social behavior, and that they can be cast out by rituals and incantations. Accordingly, they are unable to evaluate the effects that environment and experiences have upon their behavior. Many people still believe that volcanic eruptions, thunder and lightning, and all manner of cataclysms are manifestations of anger by gods or demons, and that inanimate objects have their own will and act on their own accord.

All human beings are subject to the influences of their surrounding environment. These influences become so ingrained in our habits, thoughts, feelings, and outlooks that we actually believe learned behavior is part of human nature. Even those individuals that feel they are making their own decisions, despite their cultural indoctrination, have been influenced by their surroundings. This is why we fail to critically examine our values and beliefs, and still adhere to myths, superstitions, and outdated customs which have little or no relationship to our survival.

The control of nations and individuals has not been easy because we have such little understanding about ourselves and the conditions that are responsible for shaping our behavior. People know less about their own behavior than they do about the physical world around them. That is why the dominant systems of man-made laws, and the use or threat of force have been frequently used. This technique has been tedious in its application and uncertain in its outcome. Today most of us perpetuate these conditions that may have served in earlier times but have little relevance today. What is needed is an intense research program to identify specific conditions and how those conditions influence human behavior: conditions such as environment, nutritional deficiencies, family relationships, violent-media saturation and — to a limited extent — genetic makeup.

The reason a science of human behavior has not been more widely developed is because the focus has been mostly on people and less on identifying the environmental conditions at work on the individual. The belief that our efforts should be concentrated on the development of the individual alone is fallacious. You cannot identify the factors responsible for behavior through the study of individuals, but rather through study of the cultures in which individuals are nurtured. For instance, the differences between a Native American,

a thief and a banker are not found in their genes, but instead closely parallel the environment in which they were raised.

Too many people today use genes as a scapegoat for many forms of aberrant behavior, when the major influences have been shown to be environmental. Genetic make-up alone cannot fully explain or illuminate human behavior, the science of which is a clear, if sometimes complex, algorithm of genes, environmental conditions (food, shelter, family dynamics, education, religious training, personal experiences) and the interpretation and decisions each person makes about the world and their place in it, based on these factors.

Our language causes most of our ignorance of natural law. We speak of the sun rising and setting, rather than of the earth's rotation. We talk about plants growing, inferring that this growth is of its own accord, and ignoring the relationship of growth to water, soil conditions and sunlight. When we use terms like "that rock is rolling down the hill," it implies that the rock has free will. *Nothing we have ever observed in the physical world is self-activating.* All the processes in nature are interactive. A stone does not simply roll down an inclined plane; or rivers do not simply flow. Gravity acts upon them. All living and non-living systems are acted upon by resident forces.

In like manner, the same laws that govern nature apply to human beings and are prime factors in shaping values. All human beings are immersed in an environment with an established system of values. It is the major and minor differences within that environment — and to a lesser extent the genetic attributes of the person — that are responsible for the uniqueness of the individual. If the conditions that establish those values remain unaltered, in spite of the urgings of priests, politicians, or poets, the values persist.

Perhaps in the future, if we arrive at a saner culture, people will view our notions of criminal behavior as being naive. In its very basic definition, crime is the taking of something from another without their consent. As Mark Twain once said, there is probably not an acre of land on earth that belongs to its rightful owner. Our ancestors stole the land from older peoples who took the land from others. In that sense, we are all criminals — or have at least benefited from criminal behavior.

Most man-made laws in our present culture are contrived in an attempt to control the behavior and values that serve the vested interests. If we are to reduce the crime rate, we must alter the environmental factors responsible for this socially offensive behavior. And we have to be clear about the behavior. Criminal behavior — like beauty — is often in the eye of the beholder.

In some instances crime comes about when a group of people becomes the victim of insufficient purchasing power, do not identify with the direction of society, or have little knowledge of the consequences of their actions to themselves or to the environment. In regions on earth where there is low population density and an abundance of food and water, there is no need to steal and consequently no laws against it. If the population exceeds the resources of the land then what we call criminal behavior arises as a result of scarcity,

whether artificial or real. A psychiatrist once said that if he could open a drawer and give each of his patients $200,000., 85% of his clients would have no need to see him anymore.

Today, our efforts to deal with socially offensive behavior are inadequate and inappropriate. *Eventually it will be realized and understood that most forms of so-called criminal behavior, which will fill the jails well into the twenty-first century, have been generated by the scramble for money and property in an age of often-contrived scarcity and planned obsolescence.* Four out of five of the prison population in the state of New York come from seven of the lowest income areas in that state.

Bigotry, racism, nationalism, jealousy, superstition, greed, and self-centered behavior are all learned patterns of behavior, which are strengthened or reinforced by our upbringing. These patterns of behavior are not inherited human traits or "human nature" as most people have been taught to believe. If the environment remains unaltered, similar behavior will reoccur. When we come into the world we arrive with a clean slate as far as our relationships with others are concerned.

In the final analysis, any judgment regarding undesirable human behavior serves no purpose without an attempt to alter the environment that creates it. In a society that provides for most human needs, constructive behavior would be reinforced, and people who have difficulty interacting in the community would be helped rather than imprisoned.

Aspiring to a particular ethical behavior has to do with human aspirations and ideals. *Functional morality* is the ability to provide a process level to achieve a sustainable environment for all people. By this, we mean providing clean air and water, goods and services, and a healthy and innovative environment that is emotionally and intellectually fulfilling. It is difficult to conceive of any solutions that would serve the interest of the majority in a monetary-based system. None of this can be accomplished without a comprehensive redesign of our social system and eventual replacement of the monetary system by a resource-based economy.

NATURAL LAW

Whether we realize it or not, every human being — be they criminal or saint — is a law-abiding citizen. That is to say, we are all subject to the natural laws that shape our behavior and values, and it is not possible for human life to exist without being subservient to these natural laws.

Today, however, people believe that they are independent of the laws of nature, and place themselves on a pedestal. They are unaware of their dependence upon natural laws. They build different houses of worship and plead with various interpretations of a deity to alter the laws of nature on their own behalf. They submit appeals for deliverance from such disasters as hurricanes, floods, or droughts. The world's religious leaders and their followers cannot stop outbreaks of the flu or prevent floods or hurricanes by prayer. As long as superstition and ignorance prevail, humanity will fall far short of eradicating war, poverty, and hunger. Only when the human race finally accepts the fact that they are not separate

entities in the vast symbiotic process of nature, can we truly say that there is intelligent life on earth.

Some believe that certain laws of nature — for example, the sex urge, a completely natural drive — can actually be changed by acts of Congress, and so laws are enacted against certain kinds of sexual conduct or human behavior. Such people are not swayed by the massive amounts of evidence demonstrating that these behavioral drives do not vanish even with the enactment of such laws. It is not possible to entirely prevent such behavior through legislation if it does not correlate with natural laws and principles.

Natural law is inviolable. For example, a human who does not receive proper nutrition will not enjoy physical well-being, will sicken, and will eventually die. There exist fixed properties of the physical world which no amount of human legislation can change. Natural laws are well-known, yet how many people are forced to violate them because of our social and economic insufficiencies?

With every increase in population, the values and behaviors of cultures change. With scarce resources, management and allocation of them become stringent and sets of laws evolve to correspond to the changed conditions. We must stress again that the values, habits, outlooks, beliefs, and social conduct of a given culture are determined by the environmental influences.

The earth has a built-in recycling system, an arrangement that the human race has to a large extent violated. Our rivers, oceans, and water tables overflow with debris, chemical spills, and the runoff of daily living. Landfills contain mountains of toxic and nonbiodegradable trash that will last for centuries. Replenishing the environment is very difficult in a world of unregulated competition. As fast as we recover a river, another oil platform is built in the sea. Just as all technology is engineered to perform a given task, the management of the environment that supports all life also requires an intelligent attempt to manage the output and input systems that must exist in harmony with the natural, symbiotic process.

As nations violate the symbiotic process of nature, we pay for it in the loss of arable land, environmental degradation, pollution of the oceans, territorial disputes and wars. In many instances, international agreements and laws become meaningless and counterproductive if they do not conform to the carrying capacity of the environment.

As we observe the natural world around us, we cannot help but admire the functional design and the aesthetic aspects that are byproducts of function. The ingenious economy of natural evolution has produced shapes, forms, coloration and unique configurations that appropriately conform to the environment that nurtured them.

The same laws that govern the physical world and engineering principles are universal in their application to the humanities. What distinguishes the technical person — the scientist or engineer — from the politician or theologian is that, when confronted with technical breakdowns, the former cannot blame the opposite party or the hand of the almighty. They cannot blame incompetence of the former administration. If they did so, it

is unlikely that they would ever be called upon for their services. A chemical engineer cannot avoid his or her responsibility by explaining away corrosion in transfer tubes of chemical elements. He or she is responsible for the selection of the materials used. Scientists have no way of avoiding their responsibility for problems encountered. While some people shun accountability, and justify mistakes by pointing out that to err is human, most scientists and engineers seek to minimize the probability of that error. Prior to the building of a dam or any new physical structure, for example, they conduct a great many studies in order to evaluate and uncover insufficiencies in the planning.

Unfortunately, few students learn good analytical skills. The humanities are not held up to this same scrutiny. They present vague and mystical explanations of physical phenomena. Many explanations are accepted without sufficient information or study of the subjects covered. Mystical explanations cannot serve in the practice of engineering or any other branch of the physical sciences. If we do not have sufficient information, our decisions and conclusions will be inappropriate.

Few liberal arts courses provide a foundation for the intelligent analysis required for rational thinking. What is notoriously lacking in students' education is exposure to the natural sciences and the laws responsible for all natural phenomena. In our redesign of education, we propose that intelligent analysis be a core subject in all school curricula.

It has taken many years to realize that the human being is subject to the same laws of nature that propel the planets, stars, and living and non-living systems. Setting human behavior apart from these laws is arrogant, erroneous, and dangerous.

The development of robots and artificial intelligence serves as an extension of the human body. Although seemingly disconnected from us, the cybernated world serves as an advanced and objective extension of collective thinking, and of how humans relate to one another and the world we inhabit. In fact, all the hand tools of primitive tribes and their language evolved as extensions of human attributes. This same process of extensionality is expressed in our books, architecture, mathematics and all branches of the physical sciences. This includes living and non-living systems, which are interdependent to the life process that sustains all of us.

Eventually the realization of this encompassing connectivity between living and non-living systems could enable all of us to outgrow our shallow self-centeredness as a species. This self-centeredness has prevailed within and dominated the human race for untold centuries.

As long as people and their governments remain ignorant of these basic principles, humanity will suffer the consequences. Today the management of human social systems is based upon antiquated concepts and primitive superstitions that serve national interests. We cannot achieve any real progress towards social maturation, no matter how sincere the intent of governments may be, without an understanding of these laws.

The survival of the human race depends on the recognition of these unalterable principles. If we fail to use these principles and continue trying to operate from our anthropocentric pedestal, we will be doomed to repeat the same errors over and over again.

ASSOCIATIVE MEMORY

As we explore human behavior and the influence environment has on people, one question always arises: do we really think? This is a circular question that cannot be answered until we define what we mean by "thinking." Thinking is, at its simplest, talking to oneself. The term thinking evolved as a not-wholly successful means of describing a mental process that was poorly understood at the time. Thinking is influenced by the process called associative memory. Any judgment we make, value system that we uphold, or preferences that we express are always based upon associative memory. It is essentially reflective of the environment and the experiences that we have had.

An example of associative memory would be: if we see a flower somewhat similar to a rose, but with a small, dark spot in the middle, we would probably smell the flower. After all, our experience is that roses smell pleasant. If the odor is instead pungent, the distinguishing black spot would affect our future response with other flowers of similar configuration. We may not shy away from the roses without dark spots, but we will think twice about smelling those we associate with an unpleasant experience.

Associative memory identifies objects, places or people. The same process applies to hearing, touch, smell, feelings, judgments, and opinions. All decision-making systems are based upon associative memory. This is essentially the process by which we formulate our decisions of right, wrong, good, bad, and how we measure aesthetics and beauty. Beauty lies not in the eye, but in the associative memory of the beholder.

To an entomologist the structure of a spider may be appealing, even beautiful, while others may find it repulsive. If we lived in a land where everyone had a nose six inches long, those who did not "measure up" would no doubt have surgery to add to it, to conform to the accepted norms. When an Eskimo not exposed to modern civilization thinks of transportation, it is most likely in the form of a team of dogs pulling a sled. If exposed to no other housing styles, the natives of the Amazon jungle think of a home as a thatched hut. No human being is capable of overcoming the influences of his or her environment; this includes all of one's experiences. Ample examples of this behavior exist in our own culture.

Most of us operate under the assumption that the human brain is a reservoir of limitless, untapped information. People speak of bringing out the best and noblest qualities in humans, but you cannot bring out that which is not there. This notion of the human mind is extremely dangerous and unfounded.

If an electrical engineer of eighty years ago had been handed a microchip and asked to speculate on its use, even if he were to dissect it he would have no basis for interpreting its function. The implications and understanding of associative memory can have a pro-

found effect on the way that we look at the world and ourselves. It may even raise questions as to how much freedom there is in our so-called individuality and freedom of choice.

HUMAN RELATIONS

The greatest lesson we learn may be that when human beings are free of debt, insecurity, and fear of their neighbors they become much more amiable. No one will be out to sell anyone anything or to deprive another of possessions or money. In a resource-based economy, the basis for unhealthy human aggression will be eliminated.

In addition to adequate food, clothing, shelter, heating, energy, health care and education, they must also receive a sense of acceptance, belonging, and self-worth. This new social design proposes just such an environment.

This heightened sociability will come about as we outgrow and remove the conditions that have been responsible for aberrant behavior through the introduction of a more humane and relevant value system. In a non-competitive world, through education and intelligent interaction, people will be lifted out of the mire of shallow self-centeredness that is born of the constant struggle for property, wealth and power.

Once freed of the necessity of meaningless drudgery, debt, money and possessions, people will no longer worry about many of the nagging concerns that consume so much of our attention, such as home mortgages, health care costs, fire insurance, economic recession and depression, and taxes. With the elimination of these burdens and the removal of the conditions that create feelings of envy, greed and competition, our lives would be far more meaningful, more humane and have fewer inconsistencies and conflicts.

HUMAN EMOTIONS

Many human emotions reflect environmental insufficiency, insecurity, and scarcity. Our emotions and how we express them are to a large extent determined by our culture. Here we do not refer to emotions due to physiological reactions such as physical pain, loud noises, or bright lights. By emotions, we mean *patterns of behavior that do not alleviate the problem.* A less scientific but more colorful way of describing emotion in this sense is that it resembles the racing of a car's engine at a stoplight, generating a lot of energy but taking us nowhere.

In fact, a great many human emotions are induced as strategies for achieving self-centered ends such as promotion of nationalism, salesmanship, seduction, flattery, and a myriad other forms of manipulation. They are used to gain control over the actions of others.

When a car skids on wet pavement and crashes, someone may tend to the situation as best he or she can, perhaps holding the injured person's hand until a doctor arrives. We consider such a person to be a caring and concerned individual. Rarely seen or appreciated is the engineer who comes along and adds an anti-skid surface to the pavement, thus elim-

inating the cause of the accident in the first place. This variety of caring is an emotion translated into a workable solution to eliminate the problem.

A caring society of the future will remove the conditions that are responsible for greed, envy, hate, revenge, and other undesirable human emotions. It will use technology to make certain emotions irrelevant, by getting rid of many of the problems that cause them. In a resource-based economy, when people no longer live in fear of losing their jobs or being insecure in old age, and when they have access to things that were not available to them in a monetary system, then love will not be merely a word, but actually manifested in a way of life. When humans learn to live in harmony with nature and with one another, then spirituality will be a way of life rather than just empty talk. In a more sophisticated and humane society, emotions would be harnessed and expressed by an appropriate behavior or action pattern.

When emotions are translated into a positive, constructive action pattern — when they are used to transcend the limitations of the present culture of war, poverty, and hunger that give rise to so many of these emotions — they will indeed become useful. When they are harnessed in such a way as to transcend the limitations of the present and become expressions of deed rather than simply habitual, unthinking reactions to stimuli, they will serve human beings far better.

Perhaps some day in the future, when there is peace on earth and abundance of resources available to everyone, many of the emotions that have bewildered us for centuries — emotions such as anger, despair, vengefulness, envy, and depression — will begin to abate in power, perhaps even disappear, due to the beneficial effects of our redesigned culture and environment.

CHAPTER 13

TECHNOPHOBIA IN A CYBERNATED AGE

THIS BOOK PROPOSES A PARTNERSHIP between scientific accuracy and imaginative projections, one that may lead to an age in which intelligence is no longer solely associated with human beings. In spite of fears to the contrary, potential problems do not lie between humans and machines, but rather with the limitations of the human intellect in a time of explosive technological development.

Many people fear rapid, technological developments, particularly the automated and cybernated aspects of machines replacing human beings — if not outright, perhaps, then to an extent that would deprive them of their livelihood. Many of these fears have been justified by the rapid increase in production technology requiring considerably fewer workers. This trend seems to be accelerating, and contributes to people's fears of being replaced by superior systems that do not require human participation.

The basic operating concepts of a monetary-based system exacerbate the problem, since profit is of more concern than the individual. Today, machines are not used to enhance the lives of employees by shortening the workday while increasing vacation time and purchasing power. Instead, industries use automation to benefit a select few and the shareholders. In this way the majority of people may very well occupy the role of nonessential personnel that have outlived their usefulness and are set aside, much as obsolete machines are scrapped today. It is not technology that is at fault, but the inhumane use of this technology for private profit. Humans contribute significantly to this misuse of technology by buying stock in and products produced by companies that show little concern for humans or the environment.

A few computer designers today harbor the irrational fear that the machines will eventually dominate people, since their designs are beginning to manifest human attributes. This is the unfounded fear of the technophobe. Machines actually care nothing about whether they turn out five thousand cars a month or five hundred. They merely function as they are designed to. They make no complaints as they toil in the hot sun harvesting crops and planting seeds without rest. They have no sweat glands or physical need for sleep.

It is because they do not have emotions that they will not conspire to enslave human kind. The technophobes, with their unfounded fears that computers and robots will even-

tually enslave the human race and take over the world, are simply *attributing human characteristics to machines.* They do not posses those human and animal characteristics we call feelings, which are activated by hunger, thirst, sensory stimulation, experience, and internal secretions. When a group of computers are destroyed in the presence of another computer there is no anger, resentment, or lust to "get even" on the part of the surviving computer. Many humans, particularly science fiction writers, project these characteristics into the machines of the future. Even when machines simulate emotions they are not genuine; they do not feel one way or another about any issue.

The fears that machines will increasingly regulate our lives, rob us of our natural instincts and eventually threaten our most cherished values, such as our family and spiritual beliefs, are erroneous. Even though machines may provide us with rapid transportation, prefabricated abundance, and artificial intelligence, people still harbor these fears.

Some individuals distrust a computerized society and the possible failures of the machines. They feel this technology makes us more like machines, driving us towards uniformity, resulting in the loss of individuality and that which we cherish most, freedom of choice and privacy.

In defense of machines, perhaps we would be better off if people behaved more like them. There is no question that some machines are poorly designed, but the flawed natures of human beings in high places surpass, by far, the illusion of the destructiveness of machines.

There is no evidence of machines acting against human beings of their own accord, except in naïve science fiction stories. Humans program machines and direct their use. It is not machines that are to be feared; it is the misuse and misdirection of these machines that threaten humankind. We must not forget that the bombing of cities, the use of nerve gas, prisons, death camps, and torture chambers have all been managed and operated by human beings, not machines. Even atomic weapons and guided missiles are built and directed by people. People pollute the environment — our air, oceans, and rivers. The use and sale of harmful drugs, the distortion of truth, bigotry, and racial hatred are all part of flawed human systems and false indoctrination.

Machines are not the danger: we are. As long as we fail to take responsibility for our relationship to our fellow human beings and the intelligent management of our planetary resources, we remain the greatest danger to our planet. If there would ever be a conflict between men and machines, we can be fairly certain as to who would start it! It is time we acknowledged that while there may be a moral high ground, none of us are currently standing on it. The most powerful testimony to our ignorance comes from the very scapegoats we blame for our social ills: too much technology, foreigners and minorities, "position of the planets," demonic influences, and subjective moral standards. None of these are relevant, they only serve to divert our attention away from the real problems.

Science and technology created none of our problems. Our problems arise from human abuse and misuse of other people, the environment, and technology. Downsizing is

not due to machines displacing people. In a more humane civilization such machines would be used to shorten the workday, increase the availability of goods and services, and lengthen vacation time. If we utilized new technology to raise the standard of living for everyone, then the infusion of machine technology would serve to benefit all people.

As the dangerous side effects of the misuse of technology escalate — environmental pollution, the over-exploitation of the land and sea, and the wasted resources of war and unnecessary human suffering — there is a backlash for the return to a simpler life and less technology. At the same time people call for the return to more humane values and a considerable reduction in the rate of technological development.

Those who nostalgically advocate a return to the "simple life" and "going back to the land" are misinformed and limited in their thinking. Try to imagine what would happen if we removed all of the machines in people's homes: the radio, television, computer, telephone, electric light, oven, refrigerator, and heating and cooling system. We do not see such people tossing their machines out of the house or going even one week without the use of their car. These are people who are preoccupied with wishful thinking and irrelevancies. They have always been free to give up their modern conveniences and move into a cave, if they choose to do so. But how far back does one really want to go?

These people are apparently unaware of the high rates of infant mortality, women dying in childbirth, malnutrition, and death due to infectious disease that were prevalent in earlier times. Any such regression or return to the past would be a vast waste of the human potential. Do we not instead require better means of communication, transportation, increased agricultural yield and housing for the billions of people throughout the world?

If those whose voices are raised against technology were able to turn back the clock in the name of some vague, humanitarian values, we would condemn millions of people to a state of permanent misery and unnecessary suffering.

The idea of a hand-tool economy in which humans spend most of their time providing for the bare necessities of life, devoting long hours of toil to digging wells, gathering wood for energy, hand pumping water and washing clothes in the river, leaves little or no room for the development of the individual. This holds true, also, for those in an industrial society seated on a production line who are exposed to a cycle of repetitive motions in the production of parts. We are using a very small portion of the human being in these instances and neglecting the most important aspect of being human, that which is supposed to set us apart form other animal forms — our intellect.

One of the most shameful aspects of the twentieth century is the degree of technological illiteracy affecting millions of people, despite access to the broadest range of knowledge the world has ever assembled. Even in the United States, vast numbers of people go through their day without the least idea of how a grocery scanner or a toilet works. They are only faintly aware of their dependency on dams, power plants, mass transportation, electrification and modern agricultural science for their very existence. When they turn on a light, they give little thought to the centrally controlled power grid that links the widely sep-

arated power stations by long distance transmission lines. These lines serve as the productive life force of industry, transportation and the electrification of the society.

Without access to electrical power, the telephone, air conditioning, radio, television, and computers, what we have come to accept as modern society would disappear. Without electricity the gas pump at your favorite station stops. Without refrigeration, preservation and transportation of food on a global scale would be impossible. No hospital could sustain life during surgery without machines that monitor the patient. All the leading nations of the world depend on technology for their very survival. Without modern nutritional facilities, public health would be threatened and our standard of living would be reduced to a hand-tooled economy.

In other words, it is technology that propels today's civilization. Without chemistry, agronomy, engineering, and modern health sciences the world as we know it would not exist. Humanity would be burdened by hard physical labor and longer hours of work just to maintain the bare necessities.

Because many people believe that there is too much emphasis on technology, they fail to see the humane aspects of science. It has actually been called "cold science" — admittedly with some justification in monetary-based societies, in which the tools of science are often directed primarily towards the goals of private profit and the maintaining of positions of advantage.

Many writers and literary people who perpetuate the myth of cold science express technical illiteracy and ignorance of the meaning of science. This may be due to their feeling excluded and wholly unable to grasp the real significance and sensitivity of science.

Some point out the detrimental effects of dams, irrigation canals, and nuclear power projects. They are strangely silent prior to the construction of these projects. In many cases only when the projects fail do the detractors make themselves known — but rarely with a viable solution or alternative. It is not the dams and the power projects that should be cut out; instead it is up to us to devise more effective and practical applications to harness nature and still protect the environment while helping to support human life.

There are always positive effects of all natural phenomena. Whether we view them as good or bad depends on the species affected, and on the effects on human civilization. When a volcano erupts, the dust may be spread over an extensive area, choking out many forms of life; but the lava also provides new soil and fertilization for the growth of new plants. Hurricanes spread seeds in vast areas where they would not otherwise have landed; such "pollination" originally supported the lush growth on many islands. The world of the future will involve harnessing and maximizing the forces of nature and redirecting them in constructive ways that would help support human life, while protecting the natural environment.

It is possible to build dams, canals, and power plants that offer far more than they do today, while minimizing the negative effects. For example, dams can allow for the migration of fish by means of inclined steps that enable fish to ascend to a higher level, or allow for the removal of silt where needed. If we start a project with a full "map" of the proposed

plan, not only can we see and prevent damage, but also adjust the design to accommodate the current natural processes — saving time and material. Computer simulation models already exist. Most major projects, however, are undertaken to fit special agendas, primarily business or special interests, and are totally unconcerned with the existing ecology. Much is lost in the process.

Would you trust your life to a machine? Actually, you do so every time you get into an airplane or a car. Chances are you'd much rather come into San Francisco airport in a thick fog while guided by sophisticated electronic instrumentation than by a human pilot who can't see past the nose of the plane! And how many patients in a hospital are kept alive during emergencies by machine life support systems?

As with many other things humans come into contact with, they often tend to personalize the machine. Humans can become emotionally involved with their machines, and even brought to tears or anger over them. Whole groups of people in a motion picture theater watching animated pen and ink drawings identify with the characters and laugh or weep over these nonexistent entities.

People often refer to their automobiles as their "babies". Boats are referred to as "she." Many a husband, when told by his wife or teenager "I've had an accident", inquires first how badly the car was damaged.

Since the advent of the personal computer, machines have become so ingrained in the lives of many people that not only do they depend on the computers to support their livelihoods, but they depend on them to support their mental state as well. In many instances the computers become extensions of themselves and sometimes unknowingly their best friend. Computers never argue or become indignant or jealous, neither do they react to insults. Sitting at a computer keyboard, users indulge their wildest fantasies without having to deal with the hassles of personal contact. The personal computer has become an essential and very extensional part of their lives.

The technological revolution is here to stay and will eventually, whether we support it or not, free people from the never-ending struggle for security. Computers have already invaded our schools, churches, and the highest offices of government, yet they do not intend to enslave or regiment the human race into some form of uniformity. They do not inquire into their users color, creed, sexual orientation, politics, or religion. They are in some respects kinder to us than we are to ourselves.

We need more technology, not less. But we need a new kind of application of technology. If technology were managed intelligently and with human concern, it could be used to overcome scarcity and liberate millions of human beings from the scourges of poverty and social insufficiency.

Rather than consign humanity to eternal slavery to their machines in a monetary wage system, we should allow machines to free human beings from dangerous, boring, or meaningless jobs. Far from being the threat feared by technophobes, machines could be liberators, providing us the time and the resources to help us learn what it really means to be a human being and a member of a world community.

CHAPTER 14

EDUCATION: MINDS IN THE MAKING

THE MORE INTELLIGENT OUR CHILDREN, the better our lives and the richer our culture will be. Today, each and every child using drugs and living a life without direction and purpose is a damaged life we will all have to pay for in the future. It is our children who will inherit the future. With the proper information and nurturing, they will understand the earth to be a fantastic place capable of providing more than enough for the needs of everyone.

The development of a new civilization involves not only the construction of new cities for living, but also the building of positive and caring interpersonal relationships. The young and old of this new civilization will learn to live in harmony with one another. Education plays the most important role in achieving this goal, particularly in children.

The subjects of study will be related to the direction and needs of this new, evolving culture. This new curriculum will emphasize the generalist point of view and the introduction to general science. Students will be made aware of the symbiotic relationships between people, technology, and the environment; they will have a better understanding of the evolution of cultures and the application of advanced technology to this new social design.

Our schools of tomorrow will teach children to be analytical. Students will study the interrelationships of life, rather than discrete and disassociated subject matter. The focus will be on the interrelationships of humans with the earth and with each other. Early education will emphasize understanding and cooperation.

In the redesign of education, the first questions we ask are: what end is education to serve; and in a cybernated world society how do we determine the direction of education? Some goals might be:

1. Working toward the common heritage of all the world's resources.
2. Transcending the need for all of the artificial boundaries that separate people.
3. Replacing the monetary-based economy with a resource-based world economy.

4. Reclaiming and restoring the natural environment to as near natural a condition as possible.
5. Redesigning our cities, transportation systems, and agricultural and industrial plants so that they are energy efficient, clean, and able to conveniently serve the needs of all people.
6. Gradually outgrowing the need for political methods of governance, whether at the local, national, or supranational levels, as a means of social management.
7. Sharing and applying new technologies for the benefit of all.
8. Exploring, developing and using clean, renewable energy sources such as wind, solar, geothermal, and tidal power.
9. Ultimately utilizing the highest quality products for the benefit of the world's people, while eliminating planned obsolescence.
10. Focusing on interpersonal skills that will improve relationships.
11. Requiring an environmental impact study prior to construction of any mega projects.
12. Encouraging the widest range of creativity and incentives toward constructive endeavor.
13. Assisting in stabilizing the world's population through education and voluntary birth control, in order to conform to the carrying capacity of the earth.
14. Eliminating nationalism, bigotry, and prejudice.
15. Phasing out the inclination to any type of elitism, technical or otherwise.
16. Arriving at methodologies through careful research rather than random opinions.
17. Enhancing communication so that our language is more relevant to the physical conditions of the world around us
18. Providing not only the necessities of life, but also offering challenges that stimulate the mind, while emphasizing individuality rather than uniformity.
19. Finally, preparing people intellectually and emotionally for the possible changes that lie ahead.

Ultimately, these goals determine the direction education will take. If we decide to explore the moon's surface or dig a tunnel under the sea, we must first build an organization dedicated to that goal and with the capabilities to accomplish it. To develop a civilization that can provide a higher standard of living for all people and eliminate war, poverty, and hunger, society must adopt goals that can accomplish these ends.

With the impetus of a resource-based economy, this new direction in education would stress a cooperative world enterprise in which individuality, creativity and cooperation could be the norm rather than the exception. It would be free of politics, folkways and superstitions, and encourage the widest possible range of innovative thinking.

In the schools of a unified world civilization, classrooms could provide information about human behavior and the forces that shape our culture and values. All students could have access to information without restrictions of any kind. Individual ideologies would always remain as a set of tools and an associative framework, recognized to undergo self-modification and growth with new information and experiences.

What would likely be perplexing to the citizens of the future is why there was, in the past, only one Edison, one Pasteur, one Alexander Graham Bell, one Tesla and, in general, so few others of their caliber: why was it that so few original minds managed to emerge from the billions populating our planet's history?

Try to imagine a world where thousands of such individuals live and prosper at the same time, thinking and creating to their full ability — a world in which most human beings actively participate in the improvement of the earth's conditions instead of simply toiling to make a living.

At the same time, these people of the future may find it incredible that leaders of independent nations and industries could not grasp the possibilities of a social system of cooperation rather than of competition.

We desperately need a saner mode of civilization that will no longer divide humankind. Residents of new, networked communities would be educated from birth to consider themselves planetary citizens, without sacrificing freedom and individuality to any form of totalitarianism.

SCHOOLS OF TOMORROW

Education will undergo considerable improvements. Children would be given time to explore their own interests while also participating in cooperative behavior and interaction with both other children and the environment, in hands-on experiments and tours of the natural environment, production plants and all related industries.

The learning environment would encourage actual participation on simplified levels. Younger children would plant seeds in soil, irrigate, fertilize them and record their growth, as is presently done in many schools. Actually participating in plant and animal development alters forever a child's view of nature and enhances their comprehension of the way nature works, how all its many and varied functions interrelate with each other. They will come to understand that nature is a symbiotic process and that no single thing enables a plant to grow. They would see that a plant cannot grow without radiant energy from the sun, water, and nutrients, and even the fact that gravity plays a major role in the process.

Children would understand that each individual can take an idea only so far. Others invariably add to it and improve upon it. Each contribution motivates and encourages others. Ideas grow and expand like crystals into varied and complex patterns. With this better realization of our interdependence with one another, self-centeredness gradually disappears.

Patriotism and national pride, which tend to obscure the contributions of other nations, would no longer be relevant to a new, emerging culture. The children could learn, for example, that six hundred years before Christ lived the Arabs developed the electric battery. A thousand years before the Wright brothers launched their first flying machine at Kitty Hawk, the Chinese developed man-carrying kites. A Russian named Tsiolkovsky was first to describe in detail the principals of space flight. A Frenchman, Louis Pasteur, developed an inoculation against rabies. In the sixteenth century, the Italian Leonardo da Vinci envisioned the principals of flight and designed a rudimentary form of helicopter. The Polish astronomer, Nicolaus Copernicus, published his book on the revolution of the celestial bodies. Albert Einstein, a German, gave us the theory of relativity. The contributions of all nations made our standard of living possible and enriched our lives. But we still are only at the threshold of the future.

Students would learn that no single nation has all of the answers nor an answer for all situations. Society is in a constant process of change. Students would understand that there are no final frontiers. They would also realize that each discrete phase of society will evolve a set of values appropriate to that time. All values, including many of the postulates of science, must be utilized as the best tools available at the time. With the advent of additional information and more sophisticated tools, our notion about the nature of the world could be constantly updated. Science would be taught as a set of known facts and applications that would be subject to change as more information becomes available, not as a set of immutable rules and laws.

Children taught through cooperative, participatory, hands-on experiences develop better socialization skills and self-confidence. Rather than rote learning, our new schools provide opportunities for children to improve their interaction with one another in real life situations. In other instances students may choose to explore their independent interests by selecting their own curriculum. If they prefer, they would be assisted by counselors or artificial intelligence machines, which would convey information through words, diagrams, visual displays, and many other methods. Our new schools will accommodate the many, varied ways in which children learn.

Education would emphasize human values and communication, an essential process to improve the interaction and communication between people of all races, color and creeds. Both children and adults can learn to outgrow the self-centeredness that dominates the behavior of many people today. A new form of education could make abundantly clear that our likes and dislikes are based upon our present culture, and that our visions of the future are always culture-bound.

The children would visit the farms, the power plants, the production facilities and the resource centers, and could actually take part in managing and planning their own affairs. Each child could experience leadership by planning activities, and these responsibilities could be constantly rotated so that each student would gain the experience.

If we wish to improve the mental condition of all our children, we must not only educate them through books and other visual aids, but also through games that are both physically and mentally stimulating.

The Children's Centers would be equipped with such books, computers, and a wide variety of visual aids. At these learning centers, the kind of games children play would be relevant to the needs of the child and the emergent culture. Today, far too many of the games available to our children depend on competition and encourage hostility.

The game of chess does not generate creativity in other areas; with practice, strategies for chess may be enhanced but it does not enhance creativity outside of that game. This game takes a tremendous amount of effort to learn but is about nothing in particular. If this same effort was applied to games that improved one's understanding of such areas as nutrition, health, and disease-control, it would be far more beneficial for the player and for society. People consider the chess game a challenge, but its significance is equal to that of a beauty contest. What games will be available in the future?

Consider a game centered on a virtual image of the Earth. As the children touch various areas of the Earth they could learn about the geography and languages of those areas. With laser indicators pinpointing specific areas, they could interact with and receive relevant information about any aspect of the geographical area. This could be done with just as much fun and challenge as are provided by many of the games played today, yet without the need for outsmarting the other players.

Other significant games will translate information about the physical to the needs of the individual and society. There might be games that further one's mathematical abilities. Skeletal structures of humans and other animals, when touched, would verbally identify the structures and organs, teaching anatomy and physiology. The study of plants and all other physical phenomena might be similar.

Further games will encourage creativity. In such an environment of creative games, associative memory and the experience gained form the basis of creative thinking. To think is to make a correlation, and it is the relevance of the correlation that counts. All imagination is based on cumulative experience. The broader the background, the more a person can bring to a subject.

The people of the future would be encouraged to engage in constructive diversity. Even nursery children could participate in games to develop a wide range of flexibility, individual initiative, and creativity, along with a high degree of self-sufficiency. For example, if they were told that 4 and 4 were 8, they would probably reply: "Eight what?" If two drops of fluid are suspended in an acoustical chamber we can by sonic means convert two drops to eight drops, or one drop to four drops. All numerical relationships in the future would be conducted within a given frame of reference.

Today children are not properly taught how to ask questions and examine ideas. Education consists primarily of rote learning, of simply memorizing concepts and propaganda. Children in the future would not be satisfied to accept ideas without an in-depth

exploration and understanding of them. If a child of the future were to be told that the country they lived in was the greatest in the world, they might ask "How so?" and "Compared to what set of standards?" The free minds of the twenty-first century would challenge everything — and most would, in fact, be experts at changing their minds.

At an early age children could be exposed to social and cultural anthropology. They could also be exposed to the history of civilization and to the history of technology from bow and arrow to the space age.

Rather than trying to instill in them a sense of self-worth through moral lectures, we could urge children to develop the necessary skills to further their inquiry. Education in the future could be based upon utilizing and harnessing the natural curiosity of children. The children would not, however, get instantaneous fulfillment of their requests. This tends to diminish incentive and makes it almost impossible for them to live without immediate gratification. For example, if a child asked a parent to build a model airplane the parent could say, "I will teach you how to build one". This provides the child with an appreciation for their own accomplishments and improves their self worth. As this process continues over the years the child will develop a higher self-sufficiency and depend less upon others.

Children find animated toys exciting and interesting. In the new schools they could develop their own animated toys. Before they actually build these models, they would be instructed in the use of the necessary crafts. As they develop their skills in working with these tools — while soldering, wiring, bonding, and fabricating — they can actually see and use the results. This would give them an appreciation of the effort that is required to make items they often take for granted. They could constantly learn how to apply high safety standards while working with simple and eventually more complex machines.

At the same time, the students could learn how to design and draw the models they intend to build both by hand and by computer. Science, mathematics, art, written communications, and interpersonal skills come into play in this single task. Once a project is completed, the students will have a better understanding of the relationship between the blueprint and the materials required for completing a project. Mathematics would be taught as part of the design initiative in the building of these projects, so that there is a physical reference for numerical systems. In this more advanced system, it should be an easy task to transfer these principles to all other areas of creativity within the arts and sciences. Through this process, students will be able to grasp the relationships between nature, technology, and civilization.

If we want the children to achieve a positive, constructive relationship with one another and become a contributing member of society, an effective way to accomplish this is by designing an environment that produces that desired behavior. For example, when the children are interested in learning how to assemble a small motor vehicle the design would require four children to lift the car while two others attached the wheels. The rest of the car would be assembled in a similar manner, needing the help and cooperation of everyone to

complete the vehicle for their use. This enlightened form of education would help students understand the advantages of cooperation.

Exercise in our school would not be mandatory, monotonous, or involve adversarial competition, but would be incorporated into the classroom experience. A craft shop the children enjoy using might be located on a hilltop in the middle of a lake. To get there the children would have to row a boat or swim, and then climb the hilltop. This not only provides the exercise but also a sense of achievement, which helps their mental health and incentive.

These are simplified examples of complex processes and ideas which should be considered for our redesign of education.

A great deal of attention would be given to emotional development. But perhaps the most important aspect would be on learning to interact effectively with others, share experiences, examine alternative approaches to problems, and allow for cultural and individual differences. This could reduce personal and interpersonal conflicts considerably.

Children will learn to modify their approach to get their point across, employing reason and restraint rather than name-calling or raising their voices. They would learn how to honestly disagree without experiencing bitter feelings. Such judgmental terms as "right" and "wrong" would be avoided and phased out. They would have a much more refined vocabulary and understand such terms as "a closer approximation of reality". Their vocabulary would also be a relevant one, rather than a purely emotional one. A relevant vocabulary will pertain to the situation — for example, "The inclined ramp is too steep for elderly people" — rather than an emotional vocabulary were one might remark, "A moron must have built that ramp." In other words, the child will learn that descriptive and constructive rather than outright criticism may serve to improve the situation.

Education would be participatory. Students would work cooperatively as a team. For example, if a class-group were hiking through a wooded area and came to a stream, one of the children might say to both adults and peers, "I have an idea, and I'd like to hear what you think about it." With this exposure children will have a tendency to listen and ask questions. Rather than being met with phrases such as "that will never work", students and instructors could submit their ideas to the class and test the validity of their proposals, receiving suggestions rather than criticism alone.

These young people would willingly interact with the environment, taking an active role in hiking, exploring, and investigating natural phenomena. The environment would be structured to deliver only the best in nutrition and health. Most importantly, when confronted with an unfamiliar question or situation, not only would they know where to look for appropriate information, they would know the appropriate questions to ask, and how to ask them.

Most children in our culture do not learn to describe physical processes adequately because they do not have a vocabulary equal to their physical abilities. They are not encouraged to formulate such descriptions in their daily lives; therefore they do not develop an

appropriate, descriptive language. There is an old truism that says: Once one can correctly state the problem, a solution is not far off.

When children grow up having a physical reference for the words they use, it will provide them with a much more realistic understanding of their world and of their relationship to it. Utilizing these methods, a child will gain a propensity towards problem solving that may be utilized in varying situations in the future. Rather than reacting from an emotional or uninformed standpoint, they would tend to ask, "What is the nature of the situation?" or "What do we have here?" This unique type of education will help children become more creative and participatory members of society.

Children would learn that it takes a great many experiments and a great deal of effort to solve problems. Through this process, they realize that although they may fail initially to achieve what they set out to do, this is an acceptable part of human experience. They would be taught that in medical research and other fields it sometimes takes thousands of unsuccessful experiments before arrival at a solution. Even the experiments that fail generally succeed as essential steps in the process of achieving a goal. And sometimes other discoveries emerge along the way. The children would learn not to get discouraged with these failures, that they are an inherent part of all research and development.

Few of our schoolbooks refer to the long, tedious work required to invent an object like the light bulb. No single individual manages one, great leap in technology or science without first taking several strides. Each invention is a result of a series of progressive refinements, one upon another. Every success results from the many failures and successes that have preceded it. Unfortunately, all too often our romantic notions and egos get in the way of this understanding.

This dependency on the serial progression of creativity can easily be verified if we examine the history of invention. This same process applies to the arts and sciences.

Children will come to understand that no single entity, living or nonliving, is self-activated. This concept is referred to as the mechanistic point of view. For example, a ball does not simply roll down a hill: it is acted upon by gravity. The heat from the sun is generated by a nuclear furnace; this furnace is set in motion by immense pressure. A child will ask what makes an airplane fly, as though there were a simple answer. They ask, "Is it the propeller?" No — it requires an engine to turn a propeller. They reply, "Is it the engine?" No — the engine requires fuel. They say, "Oh, is it the fuel?" In other words, there are many interacting principals at work, involving both aerodynamics and physics. All things are acted upon by resident forces, from a single cell to the cosmos in its entirety and, as noted previously, even human behavior.

Children will learn that the assumption of a beginning or an end is a fictitious assumption. This concept is a carryover from earlier civilization's attempt to account for the nature of events in the physical world with very limited information.

Our redesigned education would be free of the influence of moribund institutions, corporate or self-interests, or any form of indoctrination of a political, national or religious

nature. Similarly, the educational system would be a continuous, seamless process, the degree of each individual's curiosity enabling them to progress to the next level without grading.

Such an education would not only emphasize science and human behavior, but would also provide students with the necessary and changing professions required to maintain social and individual growth and stability in a resource-based economy. This education would enable students to engage constructively with all members of their society as well as have the ability to engage in international communication.

Some future professions would include mathematics, nanotechnology, nuclear engineering, nuclear chemistry, automation, cybernetics, systems engineering, systems analysis, remote control technology, 3-D virtual prototyping, design of plug-in components, computer-aided design and engineering, micro-machined electronic and mechanical systems, motion control, photochemical machining, ocean sciences, automated data acquisition systems, mariculture technology, simulation technology, life sciences, ecology, sociology, behavioral sciences, advanced plasma technology, industrial design, prefabrication technology, medical and bio-engineering, nutrition and health, soil enhancement systems, recycling of waste products, space science, terraforming, technology, behavioral sciences, and a host of others for which we have no name or knowledge today.

A host of other professions will disappear in a non-monetary, resource-based economy: banking, law, sales, advertising, investment brokering, real estate management, and a myriad of other occupations concerned primarily with the exclusive use of money, property and the exchange of debt.

As a student progresses from the formative stage of his/her development to the application phase, the universities and colleges of the future would not only guide students to achieve the skills relevant to an emergent society, but would also encourage them to experiment in ways that would help solve the many social problems that remain.

In the lifelong process of education, all age levels could live in cities that themselves could be designed and operated as university cities. Universities today are designed in a structured environment to provide the most advanced opportunities to facilitate education in the arts, science, music, etc. The cities in the future would be an extension of this process for fulfilling human needs. They would all serve as living universities while constantly updating information.

Much of our education today consists of a high degree of specialization, which tends to give one tunnel vision and a narrow perspective of the real interrelationships of all physical phenomena. Today it is even difficult for one schooled in sociology to communicate in depth with other members of different professions. Students of the future would be encouraged to view the world in a more holistic manner; accordingly, they would be able to converse intelligently across the various disciplines.

Children brought up in a practical, working environment of cooperation, sharing, and understanding will absorb and learn concern for their fellow human beings, recipro-

cating the warmth and love from all the people extensional to them. When the environment is intelligently and humanely managed, the system and the individual's behavior are mutual beneficiaries, each reinforcing and rewarding the other.

In a resource-base economy, children will live in a world with values far different then today's. As a result of this education and environment, they will possess a flexibility of attitude and mind that will enable them to evaluate new and different ideas. The earlier the exposure to science with human concern, the better prepared children will be to take a more appropriate place in the cybernated world of the near future. **Science and education, when devoid of a social conscience or environmental and human concern, are meaningless.**